

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers — selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions
- Architected event-driven **full-stack** TypeScript/React developer experience with observer pattern — accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Collaborated with Chrome infrastructure team to deliver design documents and code reviews at **95% test coverage**; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on Amazon Web Services (AWS) — selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy — **30% read throughput improvement** under sustained concurrency
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

PERSONAL PROJECTS

TiMoto AI — ML Serving Platform | *vLLM, gRPC, Django, PostgreSQL, AWS, Terraform* timoto.ai

- **Sole engineer** — built end-to-end **AI-powered developer platform**: vLLM inference, gRPC services, Django REST + PostgreSQL, AWS/Terraform — making complex workflows accessible via production APIs
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to shared resource acquisition in conflicting order; redesigned call sequencing eliminating circular wait — **100% evaluation success rate** at **sub-50ms p99**
- Selected **vLLM** with **PagedAttention** over naive inference to eliminate KV cache memory fragmentation under concurrent load; deployed with continuous batching — zero OOM failures at production traffic
- Architected multi-AZ ECS Fargate with Terraform Infrastructure-as-Code (IaC) and circuit breaker pattern — **99.9% uptime, 44% cost reduction** to \$40–60/month; auto-rollback on health check failure

Pulumi — Contributor | *Go, TypeScript, Infrastructure-as-Code (IaC)* github.com/pulumi/pulumi (24.4K stars)

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Linearizability, Raft/Paxos, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA (PEFT), DPO, PyTorch, LangChain, Ray, LLM-as-a-judge evaluation, Quantization (AWQ/GPTQ)

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust, Swift

Cloud & Infrastructure: Amazon Web Services (AWS) — ECS Fargate, EC2, S3, RDS; Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Google Cloud Platform (GCP), Azure

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB